

FINAL REPORT

Human Activity Recognition using Machine Learning and Deep Learning

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Abstract

Human Activity Recognition (HAR) is an important application of machine learning that enables automatic identification of daily human activities using smartphone sensor data. This project investigates the performance of traditional machine learning and deep learning models on the UCI Human Activity Recognition (HAR) dataset, which contains accelerometer and gyroscope readings collected from smartphones.

Feature engineering techniques are used to extract meaningful statistical, temporal, and frequency-domain features for traditional machine learning algorithms. Deep learning models are trained directly on sequential sensor data to learn temporal patterns automatically.

The performance of Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost, Fully Connected Neural Network (FCNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) is evaluated using Accuracy and F1-score. Experimental results show that the GRU model achieves the highest classification performance.

1. Introduction / Problem Statement

Human Activity Recognition (HAR) is the task of automatically identifying human physical activities using data collected from wearable devices or smartphone sensors. HAR has numerous applications in healthcare, fitness monitoring, rehabilitation, elderly care, and smart environments.

Modern smartphones contain built-in accelerometers and gyroscopes capable of recording body movements continuously. Machine learning techniques can analyze these sensor signals to recognize different human activities with high accuracy.

1.1 Objectives

- Classify daily human activities using smartphone sensor data.
- Compare traditional machine learning algorithms.
- Evaluate deep learning models for sequential sensor data.
- Identify the best-performing model for Human Activity Recognition.

2. Related Work

Human Activity Recognition has been widely studied using wearable sensors and smartphone-based accelerometers and gyroscopes. Traditional machine learning methods rely on manually engineered statistical and frequency-domain features, whereas deep learning methods automatically learn complex feature representations from sequential sensor data.

Recent studies demonstrate that recurrent neural networks, particularly LSTM and GRU architectures, effectively capture temporal dependencies in sensor signals, resulting in improved activity classification performance.

3. Methodology

3.1 Dataset

The project uses the UCI Human Activity Recognition (HAR) Dataset.

- Walking
- Walking Upstairs
- Walking Downstairs
- Sitting
- Standing
- Laying

3.2 Data Preprocessing

- Loading sensor data
- Handling missing values
- Removing duplicate records
- Feature scaling
- Train/validation split

3.3 Feature Engineering

Statistical Features

- Mean
- Median
- Standard Deviation
- Variance
- Minimum
- Maximum
- Skewness
- Kurtosis

Temporal Features

- Zero Crossing Rate

- Signal Energy
- Peak-to-Peak Amplitude
- Signal Magnitude Area

Frequency Features

- Spectral Centroid
- Spectral Entropy

3.4 Machine Learning Models

- Logistic Regression
- Support Vector Machine (RBF)
- Random Forest
- XGBoost

3.5 Deep Learning Models

- Fully Connected Neural Network (FCNN)
- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)

4. Experiments and Results

4.1 Dataset Description

Dataset: **UCI Human Activity Recognition Dataset**

Sensors:

- Accelerometer
- Gyroscope

Activities:

- Walking
- Walking Upstairs
- Walking Downstairs
- Sitting
- Standing
- Laying

Training Samples: **7352**

Testing Samples: **2947**

Sampling Rate: **50 Hz**

Sequence Length: **128 Time Steps**

4.2 Evaluation Metrics

Performance is evaluated using:

- Accuracy
- Precision
- Recall
- Weighted F1-score
- Confusion Matrix

4.3 Results

Model	Accuracy	F1-score
Logistic Regression	61.59%	0.606
SVM (RBF)	60.84%	0.597
Random Forest	63.90%	0.636
XGBoost	64.85%	0.646
FCNN	68.40%	0.673
RNN	36.85%	0.244
LSTM	21.75%	0.099
GRU	82.94%	0.827

Discussion

Traditional machine learning models provide reliable baseline performance using manually engineered features. Random Forest and XGBoost outperform Logistic Regression and SVM because they effectively model nonlinear relationships among features. Deep learning models automatically learn feature representations from sequential sensor data. Among these models, the GRU architecture performs best because it captures temporal dependencies while avoiding many optimization issues encountered by vanilla RNNs and LSTMs. Some confusion occurs between similar activities such as **Sitting** and **Standing**, as their sensor patterns are highly alike.

Conclusion

This project demonstrates that both traditional machine learning and deep learning techniques can effectively recognize human activities using smartphone sensor data. While feature-based machine learning algorithms achieve satisfactory performance, recurrent neural networks significantly improve activity classification by learning temporal relationships directly from sequential sensor data. Among all evaluated models, the GRU model achieved the highest classification accuracy (82.94%), making it the most effective approach for Human Activity Recognition.

Future Work

- Hyperparameter optimization
- CNN-GRU hybrid architectures
- Transformer-based activity recognition models
- Real-time activity recognition on mobile devices
- Evaluation on larger and more diverse HAR datasets